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$$a\leqslant x_0<x_1<\cdots <x_n\leqslant b$$

$$y_j=f(x_j),m_j=f'(x_j)$$

$$j=0,1,\cdots,n$$

$$H(x)$$

$$H(x_j)=y_j,\quad H'(x_j)=m_j\quad (j=0,1,\cdots,n). \tag{2.6.1}$$

$$2n+2$$

$$2n+1$$

$$H_{2n+1}(x)=H(x),$$

$$H_{2n+1}(x)=a_0+a_1x+\cdots+a_{2n+1}x^{2n+1}.$$

$$2n+2$$

$$a_0,a_1,\cdots,a_{2n+1}$$

$$\alpha_j(x)$$

$$\beta_j(x)$$

$$j=0,1,\cdots,n$$

$$2n+2$$

$$2n+1$$

$$\left\{\begin{array}{l} \alpha_j(x_k)=\delta_{jk}=\begin{cases} 0, & j\neq k, \\ 1, & j=k, \end{cases} & \alpha'_j(x_k)=0 \quad (j,k=0,1,\cdots,n), \\ \beta_j(x_k)=0, & \beta'_j(x_k)=\delta_{jk}. \end{array}\right. \tag{2.6.2}$$

$$H(x)=H_{2n+1}(x)$$

$$H_{2n+1}(x)=\sum_{j=0}^n[y_j\alpha_j(x)+m_j\beta_j(x)].\tag{2.6.3}$$

$$H_{2n+1}(x_k)=y_k,\quad H'_{2n+1}(x_k)=m_k\quad (k=0,1,\cdots,n).$$

$$\alpha_j(x)$$

$$\beta_j(x)$$

$$l_j(x)$$

$$\alpha_j(x)=(ax+b)l_j^2(x),$$

$$l_j(x)$$

$$\alpha_j(x_j)=(ax_j+b)l_j^2(x_j)=1,$$

$$\alpha_j'(x_j)=l_j(x_j)[al_j(x_j)+2(ax_j+b)l_j'(x_j)]=0,$$

$$\begin{cases} ax_j+b=1,\\ a+2l_j'(x_j)=0. \end{cases}$$

$$a=-2l_j'(x_j),\quad b=1+2x_jl_j'(x_j).$$

$$l_j(x)=\frac{(x-x_0)\cdots(x-x_{j-1})(x-x_{j+1})\cdots(x-x_n)}{(x_j-x_0)\cdots(x_j-x_{j-1})(x_j-x_{j+1})\cdots(x_j-x_n)},$$

$$l_j'(x_j)=\sum_{\substack{k=0\\k\neq j}}^n\frac{1}{x_j-x_k},$$

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$$\alpha_j(x)=\left[1-2(x-x_j)\sum_{\substack{k=0\\k\neq j}}^n\frac{1}{x_j-x_k}\right]l_j^2(x).\tag{2.6.4}$$

$$\beta_j(x)=(x-x_j)l_j^2(x).\tag{2.6.5}$$

$$H_{2n+1}(x)$$

$$\overline{H}_{2n+1}(x)$$

$$\varphi(x)=H_{2n+1}(x)-\overline{H}_{2n+1}(x)$$

$$x_k$$

$$\varphi(x)$$

$$2n+2$$

$$\varphi(x)$$

$$2n+1$$

$$\varphi(x)\equiv 0$$

$$f(x)$$

$$(a,b)$$

$$2n+2$$

$$R(x)=f(x)-H_{2n+1}(x)=\frac{f^{(2n+2)}(\xi)}{(2n+2)!}\omega_{n+1}^2(x),\tag{2.6.6}$$

$$\xi\in(a,b)$$

$$x$$

$$n=1$$

$$x_k$$

$$x_{k+1}$$

$$H_3(x)$$

$$\begin{cases} H_3(x_k)=y_k, & H_3(x_{k+1})=y_{k+1}, \\ H_3'(x_k)=m_k, & H_3'(x_{k+1})=m_{k+1}. \end{cases}\tag{2.6.7}$$

$$\alpha_k(x),\alpha_{k+1}(x),\beta_k(x),\beta_{k+1}(x)$$

$$\begin{aligned}
\alpha_k(x_k) &= 1, & \alpha_k(x_{k+1}) &= 0, & \alpha'_k(x_k) &= \alpha'_k(x_{k+1}) = 0, \\
\alpha_{k+1}(x_k) &= 0, & \alpha_{k+1}(x_{k+1}) &= 1, \\
\alpha'_{k+1}(x_k) &= \alpha'_{k+1}(x_{k+1}) = 0, & \beta_k(x_k) &= \beta_k(x_{k+1}) = 0, \\
\beta'_k(x_k) &= 1, & \beta'_k(x_{k+1}) &= 0, & \beta_{k+1}(x_k) &= \beta_{k+1}(x_{k+1}) = 0, \\
\beta'_{k+1}(x_k) &= 0, & \beta'_{k+1}(x_{k+1}) &= 1.
\end{aligned}$$

$$\begin{cases} \alpha_k(x) = \left(1 + 2\frac{x - x_k}{x_{k+1} - x_k}\right) \left(\frac{x - x_{k+1}}{x_k - x_{k+1}}\right)^2, \\ \alpha_{k+1}(x) = \left(1 + 2\frac{x - x_{k+1}}{x_k - x_{k+1}}\right) \left(\frac{x - x_k}{x_{k+1} - x_k}\right)^2; \end{cases} \quad (2.6.8)$$

$$\begin{cases} \beta_k(x) = (x - x_k) \left(\frac{x - x_{k+1}}{x_k - x_{k+1}}\right)^2, \\ \beta_{k+1}(x) = (x - x_{k+1}) \left(\frac{x - x_k}{x_{k+1} - x_k}\right)^2. \end{cases} \quad (2.6.9)$$

$$H_3(x) = y_k \alpha_k(x) + y_{k+1} \alpha_{k+1}(x) + m_k \beta_k(x) + m_{k+1} \beta_{k+1}(x), \quad (2.6.10)$$

$$R_3(x) = f(x) - H_3(x)$$

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$$R_3(x) = \frac{1}{4!} f^{(4)}(\xi) (x - x_k)^2 (x - x_{k+1})^2.$$

$$P(x_j) = f(x_j)$$

$$j = 0, 1, 2$$

$$P'(x_1) = f'(x_1)$$

$$(x_0, f(x_0)), (x_1, f(x_1))$$

$$(x_2, f(x_2))$$

$$\begin{aligned}
P(x) &= f(x_0) + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_2](x - x_0)(x - x_1) \\
&\quad + A(x - x_0)(x - x_1)(x - x_2),
\end{aligned}$$

$$A$$

$$P'(x_1) = f'(x_1)$$

$$A = \frac{f'(x_1) - f[x_0, x_1] - (x_1 - x_0)f[x_0, x_1, x_2]}{(x_1 - x_0)(x_1 - x_2)}.$$

$$R(x)=f(x)-P(x)$$

$$R(x)=f(x)-P(x)=K(x)(x-x_0)(x-x_1)^2(x-x_2),$$

$$K(x)$$

$$\varphi(t)=f(t)-P(t)-K(x)(t-x_0)(t-x_1)^2(t-x_2).$$

$$\varphi(x_j)=0$$

$$j=0,1,2$$

$$\varphi'(x_1)=0,\varphi(x)=0$$

$$\varphi(t)$$

$$(a,b)$$

$$\varphi^{(4)}(t)$$

$$(a,b)$$

$$\xi$$

$$\varphi^{(4)}(\xi)=f^{(4)}(\xi)-4!K(x)=0,$$

$$K(x)=f^{(4)}(\xi)/4!$$

$$R(x)=f^{(4)}(\xi)(x-x_0)(x-x_1)^2(x-x_2)/4!,\tag{2.6.11}$$

$$\xi$$

$$x_0,x_1,x_2$$

$$x$$

$$[a,b]$$

$$L_n(x)$$

$$f(x)$$

$$L_n(x)$$

$$n$$

$$f(x)$$

$$n\rightarrow\infty$$

$$L_n(x)$$

$$f(x)$$

$$L_n(x)$$

$$f(x)$$

$$f(x)=1/(1+x^2)$$

$$[-5,5]$$

$$[-5,5]$$

$$n+1$$

$$x_k=-5+10\frac{k}{n}\quad(k=0,1,\cdots,n)$$

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$$L_n(x)=\sum_{j=0}^n\frac{1}{1+x_j^2}\frac{\omega_{n+1}(x)}{(x-x_j)\omega'_{n+1}(x_j)}$$

$$n\rightarrow\infty$$

$$|x|\leqslant 3.63$$

$$n=10$$

$$y=L_{10}(x)$$

$$y=1/(1+x^2)$$

$$[-5,5]$$

$$x=\pm 5$$

$$L_{10}(x)$$

$$f(x)=1/(1+x^2)$$

$$L_{10}(4.8)=1.804\,38, f(4.8)=0.041\,60$$

$$L_n(x)$$

$$f(x)$$

$$y=1/(1+x^2)$$

$$x=0,\pm 1,\pm 2,\pm 3,\pm 4,\pm 5$$

$$L_{10}(x)$$

$$f(x)$$

$$x$$

$$y$$

$$O$$

$$-5,5$$

$$0.5,1.0,1.5$$

$$L_{10}(x)$$

$$\frac{1}{1+x^2}$$

$$f(x)$$

$$a = x_0 < x_1 < \cdots < x_n = b$$

$$f_0, f_1, \cdots, f_n$$

$$h_k = x_{k+1} - x_k, h = \max_k h_k$$

$$I_h(x)$$

$$I_h(x) \in C[a, b]$$

$$I_h(x_k) = f_k$$

$$k = 0, 1, \cdots, n$$

$$I_h(x)$$

$$[x_k, x_{k+1}]$$

$$I_h(x)$$

$$[x_k, x_{k+1}]$$

$$I_h(x) = \frac{x - x_{k+1}}{x_k - x_{k+1}} f_k + \frac{x - x_k}{x_{k+1} - x_k} f_{k+1} \quad (x_k \leqslant x \leqslant x_{k+1}). \tag{2.7.1}$$

$$I_h(x)$$

$$[a, b]$$

$$I_h(x) = \sum_{j=0}^n f_j l_j(x), \tag{2.7.2}$$

$$l_j(x)$$

$$l_j(x_k) = \delta_{jk}$$

$$j, k = 0, 1, \cdots, n$$

$$l_j(x) = \begin{cases} \frac{x - x_{j-1}}{x_j - x_{j-1}}, & x_{j-1} \leqslant x \leqslant x_j \quad (j = 0 \text{ 略去}), \\ \frac{x_j - x_{j+1}}{x - x_{j+1}}, & x_j \leqslant x \leqslant x_{j+1} \quad (j = n \text{ 略去}), \\ 0, & x \in [a, b], \quad x \notin [x_{j-1}, x_{j+1}]. \end{cases} \tag{2.7.3}$$

$$l_j(x)$$

$$x_j$$

$$x\in [x_k,x_{k+1}]$$

$$\frac{x_j-x_{j+1}}{x-x_{j+1}}$$

$$x=x_{j+1}$$

$$l_j(x_{j+1})=0$$

$$\frac{x-x_{j+1}}{x_j-x_{j+1}}$$

$$x_j-x_{j+1}$$

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$$1=\sum_{j=0}^nl_j(x)=l_k(x)+l_{k+1}(x),\quad f(x)=[l_k(x)+l_{k+1}(x)]f(x).$$

$$I_h(x)=f_kl_k(x)+f_{k+1}l_{k+1}(x).$$

$$l_0(x)$$

$$x_0$$

$$x_1$$

$$l_j(x)$$

$$x_{j-1},x_{j+1}$$

$$x_j$$

$$l_n(x)$$

$$x_{n-1}$$

$$x_n$$

$$l_0(x), l_j(x), l_n(x), x_0, x_1, \cdots, x_{j-1}, x_j, x_{j+1}, \cdots, x_{n-1}, x_n$$

$$\lim_{h \rightarrow 0} I_h(x) = f(x)$$

$$\begin{aligned} |f(x) - I_h(x)| &\leqslant l_k(x)|f(x) - f_k| + l_{k+1}(x)|f(x) - f_{k+1}| \\ &\leqslant [l_k(x) + l_{k+1}(x)]\omega(h_k) \\ &= \omega(h_k) \leqslant \omega(h), \end{aligned} \tag{2.7.4}$$

$$\omega(h)$$

$$f(x)$$

$$[a,b]$$

$$x', x'' \in [a,b]$$

$$|x' - x''| \leqslant h$$

$$|f(x') - f(x'')| \leqslant \omega(h),$$

$$\omega(h)$$

$$f(x)$$

$$[a,b]$$

$$f(x) \in C[a,b]$$

$$\lim_{h \rightarrow 0} \omega(h) = 0$$

$$x \in [a,b]$$

$$\max_{a \leqslant x \leqslant b} |f(x) - I_h(x)| \leqslant \omega(h)$$

$$f(x) \in C[a,b]$$

$$\lim_{h \rightarrow 0} I_h(x) = f(x)$$

$$[a,b]$$

$$I_h(x)$$

$$[a,b]$$

$$f(x)$$

$$I_h(x)$$

$$x_k$$

$$k=0,1,\cdots,n$$

$$f_k$$

$$f'_k=m_k$$

$$k=0,1,\cdots,n$$

$$I_h(x)$$

$$I_h(x)\in C^1[a,b]$$

$$C^1[a,b]$$

$$[a,b]$$

$$I_h(x_k)=f_k,I'_h(x_k)=f'_k$$

$$k=0,1,\cdots,n$$

$$I_h(x)$$

$$[x_k,x_{k+1}]$$

$$I_h(x)$$

$$[x_k,x_{k+1}]$$

$$\begin{aligned}
I_h(x) = & \left(\frac{x-x_{k+1}}{x_k-x_{k+1}}\right)^2\left(1+2\frac{x-x_k}{x_{k+1}-x_k}\right)f_k + \left(\frac{x-x_k}{x_{k+1}-x_k}\right)^2\left(1+2\frac{x-x_{k+1}}{x_k-x_{k+1}}\right)f_{k+1} \\
& + \left(\frac{x-x_{k+1}}{x_k-x_{k+1}}\right)^2(x-x_k)f'_k + \left(\frac{x-x_k}{x_{k+1}-x_k}\right)^2(x-x_{k+1})f'_{k+1}.
\end{aligned}
\tag{2.7.5}$$

$$[a,b]$$

$$\alpha_j(x)$$

$$\beta_j(x)$$

$$j=0,$$

$$\mathbf{pdf-048}$$

$$1,\cdots,n$$

$$I_h(x)$$

$$I_h(x)=\sum_{j=0}^n[f_j\alpha_j(x)+f'_j\beta_j(x)],\tag{2.7.6}$$

$$\alpha_j(x),\beta_j(x)$$

$$\alpha_j(x)=\begin{cases} \left(\frac{x-x_{j-1}}{x_j-x_{j-1}}\right)^2\left(1+2\frac{x-x_j}{x_{j-1}-x_j}\right), & x_{j-1}\leqslant x\leqslant x_j \quad (j=0 \text{ 略去}), \\ \left(\frac{x-x_{j+1}}{x_j-x_{j+1}}\right)^2\left(1+2\frac{x-x_j}{x_{j+1}-x_j}\right), & x_j\leqslant x\leqslant x_{j+1} \quad (j=n \text{ 略去}), \\ 0, & \text{其他,} \end{cases}
\tag{2.7.7}$$

$$\beta_j(x)=\begin{cases} \left(\frac{x-x_{j-1}}{x_j-x_{j-1}}\right)^2(x-x_j), & x_{j-1}\leqslant x\leqslant x_j \quad (j=0 \text{ 略去}), \\ \left(\frac{x-x_{j+1}}{x_j-x_{j+1}}\right)^2(x-x_j), & x_j\leqslant x\leqslant x_{j+1} \quad (j=n \text{ 略去}), \\ 0, & \text{其他.} \end{cases}
\tag{2.7.8}$$

$$\alpha_j(x),\beta_j(x)$$

$$x\in [x_k,x_{k+1}]$$

$$\alpha_k(x),\alpha_{k+1}(x),\beta_k(x),\beta_{k+1}(x)$$

$$I_h(x)$$

$$I_h(x) = f_k\alpha_k(x) + f_{k+1}\alpha_{k+1}(x) + f'_k\beta_k(x) + f'_{k+1}\beta_{k+1}(x) \quad (x_k \leqslant x \leqslant x_{k+1}). \tag{2.7.9}$$

$$I_h(x)$$

$$0 \leqslant \alpha_j(x) \leqslant 1, \tag{2.7.10}$$

$$\begin{cases} |\beta_k(x)| \leqslant \frac{4}{27}h_k, \\ |\beta_{k+1}(x)| \leqslant \frac{4}{27}h_k. \end{cases} \tag{2.7.11}$$

$$f(x)$$

$$f(x)$$

$$I_h(x)$$

$$f(x) = 1$$

$$\sum_{j=0}^n \alpha_j(x) = 1$$

$$x \in [x_k, x_{k+1}]$$

$$\alpha_k(x) + \alpha_{k+1}(x) = 1. \tag{2.7.12}$$

$$x \in [x_k, x_{k+1}]$$

$$\begin{aligned} |f(x) - I_h(x)| &\leqslant \alpha_k(x)|f(x) - f_k| + \alpha_{k+1}(x)|f(x) - f_{k+1}| \\ &\quad + \frac{4}{27}h_k[|f'_k| + |f'_{k+1}|] \\ &\leqslant [\alpha_k(x) + \alpha_{k+1}(x)]\omega(h) + \frac{8}{27}h \max\{|f'_k|, |f'_{k+1}|\}, \\ |f(x) - I_h(x)| &\leqslant \omega(h) + \frac{8}{27}h \max_{0 \leqslant k \leqslant n} |f'_k| \end{aligned} \tag{2.7.13}$$

$$x \in [a, b]$$

$$\omega(h)$$

$$f(x)$$

$$[a, b]$$

$$f(x) \in C[a, b]$$

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$$\lim_{h\rightarrow 0} I_h(x)=f(x).$$

$$I_h(x)$$

$$[a,b]$$

$$f(x)$$

$$S(x)\in C^2[a,b]$$

$$[x_j,x_{j+1}]$$

$$a=x_0< x_1<\cdots < x_n=b$$

$$S(x)$$

$$x_0,x_1,\cdots,x_n$$

$$x_j$$

$$y_j=f(x_j)$$

$$j=0,1,\cdots,n$$

$$S(x_j)=y_j \quad (j=0,1,\cdots,n) \tag{2.8.1}$$

$$S(x)$$

$$S(x)$$

$$[x_j,x_{j+1}]$$

$$n$$

$$4n$$

$$S(x)$$

$$[a,b]$$

$$x_j$$

$$j=1,2,\cdots,n-1$$

$$\begin{aligned} S(x_j-0) &= S(x_j+0), \quad S'(x_j-0) = S'(x_j+0), \\ S''(x_j-0) &= S''(x_j+0), \end{aligned} \tag{2.8.2}$$

$$3n-3$$

$$S(x)$$

$$4n-2$$

$$S(x)$$

$$[a,b]$$

$$a=x_0,b=x_n$$

$$\begin{cases} S'(x_0)=f'_0, \\ S'(x_n)=f'_n. \end{cases} \tag{2.8.3}$$

$$\begin{cases} S''(x_0)=f''_0, \\ S''(x_n)=f''_n. \end{cases} \tag{2.8.4}$$

$$f(x)\in C[a,b]$$

$$h\max_k|f'_k|\rightarrow 0$$

$$f\in C^1[a,b]$$

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$$S''(x_0)=S''(x_n)=0 \tag{2.8.4'}$$

$$f(x)$$

$$x_n-x_0$$

$$S(x)$$

$$\begin{cases} S(x_0+0)=S(x_n-0),\\ S'(x_0+0)=S'(x_n-0),\\ S''(x_0+0)=S''(x_n-0), \end{cases} \tag{2.8.5}$$

$$y_0=y_n$$

$$S(x)$$

$$S(x)$$

$$S'(x)$$

$$x_j$$

$$S'(x_j)=m_j$$

$$j=0,1,\cdots,n$$

$$S(x)=\sum_{j=0}^n[y_j\alpha_j(x)+m_j\beta_j(x)], \tag{2.8.6}$$

$$\alpha_j(x),\beta_j(x)$$

$$S(x)$$

$$S'(x)$$

$$[a,b]$$

$$m_j$$

$$j=0,1,\cdots,n$$

$$S''(x_j-0)=S''(x_j+0) \quad (j=1,\cdots,n-1)$$

$$m_j$$

$$S(x)$$

$$[x_j, x_{j+1}]$$

$$\begin{aligned} S(x) = & \frac{(x-x_{j+1})^2[h_j+2(x-x_j)]}{h_j^3}y_j + \frac{(x-x_j)^2[h_j+2(x_{j+1}-x)]}{h_j^3}y_{j+1} \\ & + \frac{(x-x_{j+1})^2(x-x_j)}{h_j^2}m_j + \frac{(x-x_j)^2(x-x_{j+1})}{h_j^2}m_{j+1}, \end{aligned} \tag{2.8.7}$$

$$h_j = x_{j+1} - x_j$$

$$S(x)$$

$$\begin{aligned} S''(x) = & \frac{6x-2x_j-4x_{j+1}}{h_j^2}m_j + \frac{6x-4x_j-2x_{j+1}}{h_j^2}m_{j+1} \\ & + \frac{6(x_j+x_{j+1}-2x)}{h_j^3}(y_{j+1}-y_j), \end{aligned}$$

$$S''(x_j+0) = -\frac{4}{h_j}m_j - \frac{2}{h_j}m_{j+1} + \frac{6}{h_j^2}(y_{j+1}-y_j).$$

$$S''(x)$$

$$[x_{j-1}, x_j]$$

$$\begin{aligned} S''(x) = & \frac{6x-2x_{j-1}-4x_j}{h_{j-1}^2}m_{j-1} + \frac{6x-4x_{j-1}-2x_j}{h_{j-1}^2}m_j \\ & + \frac{6(x_{j-1}+x_j-2x)}{h_{j-1}^3}(y_j-y_{j-1}) \end{aligned}$$

$$h_{j-1}^2$$

$$S''(x)$$

$$h_{j-1}^3$$

$$x = x_j$$

$$-6(y_j-y_{j-1})/h_{j-1}^2$$

$$j-1$$

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$$S''(x_j-0)=\frac{2}{h_{j-1}}m_{j-1}+\frac{4}{h_{j-1}}m_j-\frac{6}{h_{j-1}^2}(y_j-y_{j-1}).$$

$$S''(x_j+0)=S''(x_j-0)$$

$$j=1,2,\cdots,n-1$$

$$\begin{aligned} \frac{1}{h_{j-1}}m_{j-1}+2\left(\frac{1}{h_{j-1}}+\frac{1}{h_j}\right)m_j+\frac{1}{h_j}m_{j+1} \\ =3\left(\frac{y_{j+1}-y_j}{h_j^2}+\frac{y_j-y_{j-1}}{h_{j-1}^2}\right) \quad (j=1,2,\cdots,n-1), \\ \frac{1}{h_{j-1}}+\frac{1}{h_j} \end{aligned} \tag{2.8.8}$$

$$y_j=f_j,\frac{y_{j+1}-y_j}{h_j}=f[x_j,x_{j+1}]$$

$$\lambda_jm_{j-1}+2m_j+\mu_jm_{j+1}=g_j \quad (j=1,2,\cdots,n-1), \tag{2.8.9}$$

$$\lambda_j=\frac{h_j}{h_{j-1}+h_j}, \quad \mu_j=\frac{h_{j-1}}{h_{j-1}+h_j} \quad (j=1,2,\cdots,n-1), \tag{2.8.10}$$

$$g_j=3(\lambda_jf[x_{j-1},x_j]+\mu_jf[x_j,x_{j+1}]) \quad (j=1,2,\cdots,n-1). \tag{2.8.11}$$

$$m_0,m_1,\cdots,m_n$$

$$n-1$$

$$m_0=f'_0$$

$$m_n=f'_n$$

$$m_1,\cdots,m_{n-1}$$

$$n-1$$

$$\begin{pmatrix} 2 & \mu_1 & 0 & \cdots & \cdots & 0 \\ \lambda_2 & 2 & \mu_2 & \ddots & & \vdots \\ 0 & \lambda_3 & 2 & \mu_3 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & \ddots & 0 \\ 0 & & \ddots & \lambda_{n-2} & 2 & \mu_{n-2} \\ 0 & \cdots & \cdots & 0 & \lambda_{n-1} & 2 \end{pmatrix} \begin{pmatrix} m_1 \\ m_2 \\ m_3 \\ \vdots \\ m_{n-2} \\ m_{n-1} \end{pmatrix} = \begin{pmatrix} g_1-\lambda_1f'_0 \\ g_2 \\ g_3 \\ \vdots \\ g_{n-2} \\ g_{n-1}-\mu_{n-1}f'_n \end{pmatrix}. \tag{2.8.12}$$

$$\begin{cases} 2m_0+m_1=3f[x_0,x_1]-\frac{h_0}{2}f_0''=g_0, \\ m_{n-1}+2m_n=3f[x_{n-1},x_n]+\frac{h_{n-1}}{2}f_n''=g_n. \end{cases}$$

(2.8.13)

$$\begin{cases} 2m_0+m_1=3f[x_0,x_1]=g_0, \\ m_{n-1}+2m_n=3f[x_{n-1},x_n]=g_n. \end{cases}$$

(2.8.13')

$$\begin{pmatrix} 2 & 1 & 0 & \cdots & \cdots & 0 \\ \lambda_1 & 2 & \mu_1 & \ddots & & \vdots \\ 0 & \lambda_2 & 2 & \mu_2 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & \ddots & 0 \\ \vdots & & \ddots & \lambda_{n-1} & 2 & \mu_{n-1} \\ 0 & \cdots & \cdots & 0 & 1 & 2 \end{pmatrix} \begin{pmatrix} m_0 \\ m_1 \\ m_2 \\ \vdots \\ m_{n-1} \\ m_n \end{pmatrix} = \begin{pmatrix} g_0 \\ g_1 \\ g_2 \\ \vdots \\ g_{n-1} \\ g_n \end{pmatrix}.$$

(2.8.14)

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$$m_0=m_n,$$

$$\frac{1}{h_0}m_1+\frac{1}{h_{n-1}}m_{n-1}+2\left(\frac{1}{h_0}+\frac{1}{h_{n-1}}\right)m_n=\frac{3}{h_0}f[x_0,x_1]+\frac{3}{h_{n-1}}f[x_{n-1},x_n],$$

$$\mu_nm_1+\lambda_nm_{n-1}+2m_n=g_n,$$

$$\mu_n=\frac{h_{n-1}}{h_0+h_{n-1}},\quad \lambda_n=\frac{h_{n-1}}{h_0+h_{n-1}},$$

$$g_n=3(\mu_nf[x_0,x_1]+\lambda_nf[x_{n-1},x_n]).$$

$$\begin{pmatrix} 2 & \mu_1 & 0 & \cdots & 0 \\ \lambda_2 & 2 & \mu_2 & \ddots & \vdots \\ 0 & \lambda_3 & \ddots & \ddots & 0 \\ \vdots & \ddots & \ddots & 2 & \mu_{n-1} \\ 0 & \cdots & 0 & \lambda_n & 2 \end{pmatrix} \begin{pmatrix} m_1 \\ m_2 \\ \vdots \\ m_{n-1} \\ m_n \end{pmatrix} = \begin{pmatrix} g_1 \\ g_2 \\ \vdots \\ g_{n-1} \\ g_n \end{pmatrix}.$$

(2.8.15)

$$m_j$$

$$m_j$$

$$x_j$$

$$\mu_j+\lambda_j=1$$

$$m_j$$

$$j=0,1,\cdots,n$$

$$S(x)$$

$$S(x)$$

$$S''(x_j)=M_j$$

$$j=0,1,\cdots,n$$

$$M_j$$

$$x_j$$

$$S(x)$$

$$[x_j,x_{j+1}]$$

$$S''(x)$$

$$[x_j,x_{j+1}]$$

$$S''(x)=M_j\frac{x_{j+1}-x}{h_j}+M_{j+1}\frac{x-x_j}{h_j}.$$

(2.8.16)

$$S''(x)$$

$$S(x_j)=y_j$$

$$S(x_{j+1})=y_{j+1}$$

$$\begin{aligned} S(x) &= M_j\frac{(x_{j+1}-x)^3}{6h_j}+M_{j+1}\frac{(x-x_j)^3}{6h_j}+\left(y_j-\frac{M_jh_j^2}{6}\right)\frac{x_{j+1}-x}{h_j} \\ &\quad +\left(y_{j+1}-\frac{M_{j+1}h_j^2}{6}\right)\frac{x-x_j}{h_j} \quad (j=0,1,\cdots,n-1). \end{aligned}$$

(2.8.17)

$$S(x)$$

$$S'(x)=-M_j\frac{(x_{j+1}-x)^2}{2h_j}+M_{j+1}\frac{(x-x_j)^2}{2h_j}+\frac{y_{j+1}-y_j}{h_j}-\frac{M_{j+1}-M_j}{6}h_j.$$

(2.8.18)

$$\lambda_n$$

$$\mu_n$$

$$h_{n-1}/(h_0+h_{n-1})$$

$$m_{n-1}$$

$$\lambda_n=h_0/(h_0+h_{n-1})$$

$$0$$

$$m_0=m_n$$

$$\lambda_1$$

$$\mu_n$$

$$\mathbf{pdf-053}$$

$$S'(x_j+0)=-\frac{h_j}{3}M_j-\frac{h_j}{6}M_{j+1}+\frac{y_{j+1}-y_j}{h_j}.$$

$$S(x)$$

$$[x_{j-1},x_j]$$

$$S'(x_j-0)=\frac{h_{j-1}}{6}M_{j-1}+\frac{h_{j-1}}{3}M_j+\frac{y_j-y_{j-1}}{h_{j-1}},$$

$$S'(x_j+0)=S'(x_j-0)$$

$$\mu_jM_{j-1}+2M_j+\lambda_jM_{j+1}=d_j\quad (j=1,2,\cdots,n-1),\tag{2.8.19}$$

$$\mu_j,\lambda_j$$

$$d_j=6\frac{f[x_j,x_{j+1}]-f[x_{j-1},x_j]}{h_{j-1}+h_j}=6f[x_{j-1},x_j,x_{j+1}]\quad (j=1,2,\cdots,n-1).\tag{2.8.20}$$

$$M_j$$

$$2M_0+M_1=\frac{6}{h_0}(f[x_0,x_1]-f'_0),\quad M_{n-1}+2M_n=\frac{6}{h_{n-1}}(f'_n-f[x_{n-1},x_n]).$$

$$M_0=f''_0,\quad M_n=f''_n.$$

$$M_j$$

$$j=0,1,\cdots,n$$

$$S(x)$$

$$S(x)$$

$$x_j,y_j$$

$$j=0,1,\cdots,n$$

$$f_0',f_n'$$

$$n$$

$$j$$

$$n-1$$

$$h_j=x_{j+1}-x_j$$

$$f[x_j,x_{j+1}]$$

$$j$$

$$n-1$$

$$\lambda_j,\mu_j,g_j$$

$$m_j$$

$$j=1,2,\cdots,n-1$$

$$S(x)$$

$$S(x)$$

$$f(x)=\frac{1}{1+x^2},-5\leqslant x\leqslant 5$$

$$x_k = -5 + k$$

$$k = 0, 1, \cdots, 10$$

$$S_{10}(x)$$

$$S_{10}(x_k) = f(x_k)$$

$$k = 0, 1, \cdots, 10$$

$$S'_{10}(-5) = f'(-5), \quad S'_{10}(5) = f'(5).$$

$$S_{10}(x)$$

$$f(x)$$

$$L_{10}(x)$$

$$S_{10}(x)$$

$$f(x)$$

$$L_{10}(x)$$

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$$f(x)$$

$$L_{10}(x)$$

$$S_{10}(x)$$

$$f(x)$$

$$x$$

$$\frac{1}{1+x^2}$$

$$S_{10}(x)$$

$$L_{10}(x)$$

$$x$$

$$\frac{1}{1+x^2}$$

$$S_{10}(x)$$

$$L_{10}(x)$$

$$\boldsymbol{A} = (a_{ij})_n$$

$$n \times n$$

$$\boldsymbol{x} = (x_1,x_2,\cdots,x_n)^{\mathrm{T}}$$

$$n$$

$$\boldsymbol{x}$$

$$\boldsymbol{A}$$

$$\begin{aligned}\|\boldsymbol{x}\|_\infty &= \max_{1\leqslant i\leqslant n}|x_j|, \\ \|\boldsymbol{A}\|_\infty &= \sup_{\boldsymbol{x}\neq 0}\frac{\|\boldsymbol{A}\boldsymbol{x}\|_\infty}{\|\boldsymbol{x}\|_\infty}.\end{aligned}\tag{2.8.21}$$

$$f(x)\in C[a,b]$$

$$f$$

$$\|f\|_\infty=\sup_{a\leqslant x\leqslant b}|f(x)|=\max_{a\leqslant x\leqslant b}|f(x)|.$$

$$\boldsymbol{A} = (a_{ij})_n$$

$$\sum_{j\neq i}|a_{ij}|<|a_{ii}| \quad (i=1,2,\cdots,n),\tag{2.8.22}$$

$$\boldsymbol{A}^{-1}$$

$$\|\boldsymbol{A}^{-1}\|_\infty\leqslant\left\{\min\left(|a_{ij}|-\sum_{j\neq i}|a_{ij}|\right)\right\}^{-1}.\tag{2.8.23}$$

$$S(x)$$

$$\max$$

$$i$$

$$x_j$$

$$|a_{ij}|$$

$$|a_{ii}|$$

$$i$$

$$S_{10}$$

$$f(x)=1/(1+x^2)$$

$$S'_{10}(\pm 5)=f'(\pm 5)$$

$$S_{10}(-4.8)\approx 0.04162183$$

$$S_{10}(-4.5)\approx 0.04716801$$

$$0.03758$$

$$0.04248$$

pdf-055

$$\boldsymbol{A}\boldsymbol{m}=\boldsymbol{g}, \tag{2.8.24}$$

$$\boldsymbol{A}=\begin{pmatrix} 2 & 1 & 0 & \cdots & 0 \\ \lambda_1 & 2 & \mu_1 & \ddots & \vdots \\ 0 & \lambda_2 & \ddots & \ddots & 0 \\ \vdots & \ddots & \ddots & 2 & \mu_{n-1} \\ 0 & \cdots & 0 & \lambda_n & 2 \end{pmatrix}, \quad \boldsymbol{m}=\begin{pmatrix} m_0 \\ m_2 \\ \vdots \\ m_{n-1} \\ m_n \end{pmatrix}, \quad \boldsymbol{g}=\begin{pmatrix} g_0 \\ g_1 \\ \vdots \\ g_{n-1} \\ g_n \end{pmatrix}.$$

$$\mu_i+\lambda_i=1$$

$$i=0,1,\cdots,n$$

$$a_{ii}=2$$

$$\| \boldsymbol{A}^{-1} \| \leqslant 1. \tag{2.8.25}$$

$$f(x)\in C[a,b]$$

$$S(x)$$

$$a=x_0<x_1<\cdots<x_n=b$$

$$h_j=x_{j+1}-x_j,\quad h=\max_{0\leqslant j\leqslant n-1}h_j,\quad \delta=\min_{0\leqslant j\leqslant n-1}h_j,$$

$$h/\delta\leqslant c<\infty$$

$$S(x)$$

$$[a,b]$$

$$f(x)$$

$$S(x)$$

$$x\in[a,b]$$

$$\|f(x)-S(x)\|_\infty\leqslant\omega(h)+(8/27)h\|\boldsymbol{m}\|_\infty, \tag{2.8.26}$$

$$\|\boldsymbol{m}\|_\infty\leqslant\|\boldsymbol{A}^{-1}\|_\infty\|\boldsymbol{g}\|_\infty\leqslant\|\boldsymbol{g}\|_\infty^m, \tag{2.8.27}$$

$$\|\boldsymbol{g}\|_\infty\leqslant3\max_{0\leqslant j\leqslant n-1}|f[x_j,x_{j+1}]|\leqslant\frac{3}{\delta}\omega(h)\leqslant\frac{6}{\delta}\|f\|_\infty, \tag{2.8.28}$$

$$\|f(x)-S(x)\|_\infty\leqslant\left(1+\frac{8}{9}\frac{h}{\delta}\right)\omega(h).$$

$$f(x)\in C[a,b]$$

$$h\rightarrow0$$

$$\|f(x)-S(x)\|_\infty\rightarrow0$$

$$S(x)$$

$$[a,b]$$

$$f(x)$$

$$f(x)\in C^1[a,b]$$

$$S'(x)$$

$$[a,b]$$

$$f'(x)$$

$$B$$

$$\lambda_n$$

$$1$$

$$\boldsymbol{m}$$

$$m_2$$

$$m_1$$

$$\lambda_n$$

$$m$$

$$\|\boldsymbol{g}\|_\infty^m$$

$$\|\boldsymbol{m}\|_\infty\leqslant\|\boldsymbol{g}\|_\infty$$

pdf-056

$$V_n(x)=V_n(x_0,x_1,\cdots,x_{n-1},x)=\begin{vmatrix} 1&x_0&x_0^2&\cdots&x_0^n\\ \vdots&\vdots&\vdots&&\vdots\\ 1&x_{n-1}&x_{n-1}^2&\cdots&x_{n-1}^n\\ 1&x&x^2&\cdots&x^n \end{vmatrix},$$

$$V_n(x)$$

$$n$$

$$x_0,x_1,\cdots,x_{n-1}$$

$$V_n(x)=V_{n-1}(x_0,x_1,\cdots,x_{n-1})(x-x_0)\cdots(x-x_{n-1}).$$

$$x=1,-1,2$$

$$f(x)=0,-3,4$$

$$f(x)$$

$$f(x)=\ln x$$

$$\ln 0.54$$

$$x$$

$$\ln x$$

$$\cos x, 0^{\circ} \leqslant x \leqslant 90^{\circ}$$

$$h=1'=(1/60)^{\circ}$$

$$\cos x$$

$$x_k=x_0+kh,k=0,1,2,3$$

$$\max_{x_0\leqslant x\leqslant x_3}|l_2(x)|$$

$$x_j$$

$$j=0,1,\cdots,n$$

$$\sum_{j=0}^n x_j^k l_j(x) \equiv x^k$$

$$k=0,1,\cdots,n$$

$$\sum_{j=0}^n (x_j-x)^k l_j(x) \equiv 0$$

$$k=1,2,\cdots,n$$

$$f(x)\in C^2[a,b]$$

$$f(a)=f(b)=0$$

$$\max_{a\leqq x\leqq b}|f(x)|\leqq \frac{1}{8}(b-a)^2\max_{a\leqq x\leqq b}|f''(x)|$$

$$-4\leqq x\leqq 4$$

$$f(x)=e^x$$

$$e^x$$

$$10^{-6}$$

$$h$$

$$y_n=2^n$$

$$\Delta^4 y_n$$

$$\delta^4 y_n$$

$$f(x)$$

$$m$$

$$\Delta f(x)=f(x+h)-f(x)$$

$$f(x)$$

$$k$$

$$\Delta^kf(x)$$

$$0\leqslant k\leqslant m$$

$$m-k$$

$$\Delta^{m+l}f(x)=0$$

$$l$$

$$\Delta(f_k g_k)=f_k\Delta g_k+g_{k+1}\Delta f_k$$

$$\sum_{k=0}^{n-1}f_k\Delta g_k=f_ng_n-f_0g_0-\sum_{k=0}^{n-1}g_{k+1}\Delta f_k$$

$$\sum_{j=0}^{n-1}\Delta^2y_j=\Delta y_n-\Delta y_0$$

$$f(x)=a_0+a_1x+\cdots+a_{n-1}x^{n-1}+a_nx^n$$

$$n$$

$$x_1,x_2,\cdots,x_n$$

$$x=0.7$$

$$-0.357765$$

$$\ln 0.7\approx -0.356674944$$

$$-0.356675$$

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$$\sum_{j=1}^n\frac{x_j^k}{f'(x_j)}=\begin{cases}0,&0\leqslant k\leqslant n-2,\\a_n^{-1},&k=n-1.\end{cases}$$

$$n$$

$$F(x)=cf(x)$$

$$F[x_0,x_1,\cdots,x_n]=cf[x_0,x_1,\cdots,x_n]$$

$$F(x)=f(x)+g(x)$$

$$F[x_0,x_1,\cdots,x_n]=f[x_0,x_1,\cdots,x_n]+g[x_0,x_1,\cdots,x_n]$$

$$f(x)=x^7+x^4+3x+1$$

$$f[2^0,2^1,\cdots,2^7]$$

$$f[2^0,2^1,\cdots,2^8]$$

$$R_3(x)=f^{(4)}(\xi)(x-x_k)^2(x-x_{k+1})^2/4!,\quad \xi\in (x_k,x_{k+1}),$$

$$P(x)$$

$$P(0)=P(-k+1)$$

$$P(x)$$

$$P(0)=P'(0)=0,P(1)=P'(1)=1,P(2)=1$$

$$f(x)\in C[a,b]$$

$$[a,b]$$

$$n$$

$$\varphi_n(x)$$

$$n\rightarrow\infty$$

$$\varphi_n(x)$$

$$[a,b]$$

$$f(x)$$

$$f(x)=1/(1+x^2)$$

$$-5\leqslant x\leqslant 5$$

$$n=10$$

$$I_h(x)$$

$$I_h(x)$$

$$f(x)$$

$$f(x)=x^2$$

$$[a,b]$$

$$I_h(x)$$

$$f(x)=x^4$$

$$[a,b]$$

$$S(x)$$

$$S'(0.25)=1.000\,0, S'(0.53)=0.686\,8$$

$$S''(0.25)=S''(0.53)=0$$

$$x_j$$

$$y_j$$

$$f(x)\in C^2[a,b]$$

$$S(x)$$

$$\begin{aligned} &\int_a^b [f''(x)]^2 \, \mathrm{d}x - \int_a^b [S''(x)]^2 \, \mathrm{d}x \\ &= \int_a^b [f''(x) - S''(x)]^2 \, \mathrm{d}x + 2 \int_a^b S''(x)[f''(x) - S''(x)] \, \mathrm{d}x; \end{aligned}$$

$$f(x_i)=S(x_i)$$

$$i=0,1,\cdots,n$$

$$x_i$$

$$a=x_0< x_1<\cdots < x_n=b$$

$$\int_a^b S'''(x)[f''(x)-S''(x)]\,\mathrm{d}x=S'''(b)[f'(b)-S'(b)]-S'''(a)[f'(a)-S'(a)].$$

$$S(x)$$

$$S(x)$$

$$k$$